

# M50 — Silence + Clear-bugfix + BP thresholds + danger severity + Nurse-station Code Blue

**Phase:** Phase 7 follow-on (post-M49, operator feedback) **Status:** **DONE** **Blocked by:** M26 (alarm bus), M27 (Nursing Station), M48 (alarm thresholds), M49 (alarm sounds) **Blocks:** none **Estimated effort:** 1 day

## 1. Purpose

Five interrelated asks from the operator:

*"Add silence alarm for each Patient encounter on nurses station alarm in addition to clear. The alarm clear did not work. Also add Blood pressure to the list of items that has alarm ranges. For Dangerous wave forms they should get a weighted with highest alarm rating. Also the nurse station should be able to activate a code blue from the nurses station screen on a specific patient character encounter."*

Delivered:

1. **Silence button** per alarm — POST `/api/alarm/{id}/silence?seconds=N` defaults to 120 s. The alarm stays **VISIBLE** on the board with a ■ badge but no audio fires during the silence window. Works on every alarm source (device, scene, threshold).
2. **Clear bugfix** — pre-M50, `clear_alarm()` returned `None` for `source=threshold` (no event log to write into), so the `/api/alarm/{id}/clear` route 404'd on every threshold alarm. Fixed: threshold clears now route through the silenced map with `cleared=True` so they're filtered out of the active feed.
3. **BP thresholds** — new `bp_systolic` and `bp_diastolic` keys on `room.alarm_thresholds` with sensible adult-norm defaults (sbp 90–160, dbp 60–100). The threshold check raises alarms when either side breaches.
4. **Danger severity** — dangerous waveforms (v-fib, asystole, v-tach) now use `severity="danger"` (rank 4) which outranks `critical` (rank 3) in the alarm-board sort. They float to the **TOP** of the alarm feed with a red pulsing visual treatment. `alarm_sounds.severity_to_priority` maps `danger` → `high` so they share the **HIGH**-priority WAV with critical alarms.
5. **Nurse-station Code Blue** — new route POST `/api/room/encounter/{eid}/nurse_code_blue` accepting `nurse_sid` in the body (validates against the room's nurse-station students). The Nursing Station bed cards get a ■ **Code Blue** button that POSTs to this route.

## 2. Structure

**Files touched:**

- `portal/control_room.py`:
- `alarm_thresholds` defaults now include `bp_systolic` + `bp_diastolic` (adult-norm ranges). - New `silenced_alarms`: `dict[str, dict[str, Any]]` field. Maps `alarm_id` → `{until: float, cleared: bool}`. The `alarms.py_apply_silenced` helper reads + cleans up.
- `portal/alarms.py`:
- `active_alarms(room)` `severity_rank` gains a `danger=4` entry

(above critical=3) so dangerous-rhythm alarms sort to the top. - New `_apply_silenced(room, alarms)` helper: drops cleared alarms from the active feed, annotates silenced alarms with `silenced=True + silenced_until`. Cleans up expired entries. - `_threshold_alarms_for` extends `metric_breaches` to include `bp_systolic + bp_diastolic` (mapped to snapshot keys `sbp / dbp`). - Dangerous-rhythm alarms now emit `severity="danger"`. - `clear_alarm()` gets a source `== "threshold"` branch that stores the alarm in `room.silenced_alarms` with `cleared=True` and an until 24 h out (effectively "cleared until breach resolves"). - New `silence_alarm(room, alarm_id, duration_s=120)` helper — stores with `cleared=False`.

- `portal/alarm_sounds.py`:

- `severity_to_priority` now maps both danger AND critical

→ "high" (same WAV bucket — no separate danger asset ships). - `audio_url_for` collapses `bp_systolic / bp_diastolic` families into the `hr` audio family (asset library doesn't ship a dedicated BP WAV).

- `portal/server.py`:

- New POST `/api/alarm/{id}/silence?seconds=N` route (instructor gated). - New POST `/api/room/encounter/{eid}/nurse_code_blue` route accepting EITHER an instructor cookie OR `nurse_sid` body field (validates against `room.students` whose `role == "nurse_station"`). Fires a `code.blue` scene via the existing `scenes.apply()` helper. - POST `/api/room/alarm_thresholds` validation expanded to accept `bp_systolic + bp_diastolic` keys.

- `portal/templates/nurse_station.html`:

- Threshold form gets two new rows: BP systolic + BP diastolic (each with low + high inputs).

- `portal/static/nurse_station.js`:

- `renderAlarmBoard` renders BOTH a  Silence and a Clear button

per alarm; silenced alarms render with `.silenced` CSS class and a badge. - `_playNewAlarmSounds` skips alarms with `a.silenced === true`. - `loadThresholds + saveThresholds` read/write the BP fields. - `renderBedCard` adds a  Code Blue button at the bottom of each bed card with a `confirm()` dialog and POST to the new `nurse_code_blue` route with the student's `sid`.

- `portal/static/nurse_station.css`:

- `.alarm-actions` flex for the Silence + Clear button pair.

- `.silenced-badge` styling, plus `.alarm-list li.silenced` grey-

out treatment. - `.severity-danger` deep-red background + pulsing border keyframe. -

`.bed-card-actions + .ns-code-blue-btn` (deep-red CTA).

### 3. Uses

#### 3.1 Silence flow

1. Operator on Nursing Station sees a SpO2 critical alarm. 2. Clicks  **Silence**. POST `/api/alarm/threshold:...:spo2/silence` → server stores `{until: now+120, cleared: false}` in `room.silenced_alarms`. 3. Next 3 s poll: alarm still in feed, but now with `silenced=true + silenced_until=...` JS dispatcher's `_seenAlarmIds` check additionally skips audio for silenced alarms. 4. After 120 s the `until` timestamp passes; the `_apply_silenced` helper cleans up the entry on the next

read; alarm re-fires audio if still breaching.

### 3.2 Clear bugfix

| Before M50                                                                                                               | After M50                                                                                                                                                              |
|--------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Operator clicks Clear on a threshold alarm. POST 404s "Unknown alarm id 'threshold:...:spo2'". Alarm stays on the board. | Same click — route returns 200 with {cleared: true, source: "threshold"}. Alarm gone from the next read. If the breach resolves, the cleared entry expires harmlessly. |

### 3.3 BP thresholds

1. Operator opens Nursing Station threshold settings. 2. Sees BP systolic (90–160) + BP diastolic (60–100) pre-filled. 3. Adjusts to scenario-appropriate values (e.g. sbp 100–140). 4. Click Save → POST `/api/room/alarm_thresholds` body {bp\_systolic: {low: 100, high: 140}, ...}. 5. When a scene drops sbp to 75, the alarm bus raises a `threshold:E-XYZ:bp_systolic` warning alarm with label `"BP systolic low (75 < 100)"`.

### 3.4 Danger severity sorting

Two alarms simultaneously active on the room:

- v-fib rhythm on Bed 2 → `severity=danger`
- SpO2 84 on Bed 1 → `severity=critical`

The alarm board sorts:

1. ■ v-fib rhythm (danger, Bed 2) ← top, red pulsing
2. ■■ SpO2 low (critical, Bed 1)

Audio: both fire the high-priority WAV (file 04 for ECG and file 06 for SpO2). The visual treatment makes the danger alarm unmissable.

### 3.5 Nurse-station Code Blue

1. Supervisor sees Bed 2 deteriorating (v-fib + low SpO2). 2. Clicks ■ **Code Blue** on Bed 2's card. 3. Confirms in the `confirm()` dialog. 4. JS POSTs `/api/room/encounter/E-bed2/nurse_code_blue` with {nurse\_sid: "S-instructor-nurse-sid"}. 5. Server validates the sid against `room.students` (must be a `nurse_station-role` student); calls `scenes.apply(enc_bed2, {kind: "code.blue", by="nurse_station:<sid>"})`. 6. The `code.blue` scene writes chart events + raises alarms normally — same path as if the instructor had clicked "Inject scene" on the Multi-Patient Control dashboard.

Auth alternatives:

- Instructor cookie: works without `nurse_sid`.
- `nurse_sid` not matching a nurse-station student: 403.

## 4. Functions (exported API surface)

| Symbol                                                      | Where                               | Purpose                                                         |
|-------------------------------------------------------------|-------------------------------------|-----------------------------------------------------------------|
| POST <code>/api/alarm/{id}/silence?seconds=N</code>         | <code>portal/server.py</code>       | Silence an alarm for N seconds.                                 |
| POST <code>/api/room/encounter/{eid}/nurse_code_blue</code> | same                                | Fire <code>code.blue</code> from Nursing Station OR instructor. |
| <code>ControlRoom.silenced_alarms</code>                    | <code>portal/control_room.py</code> | In-memory dict of silenced/cleared alarm states.                |

| Symbol                                                                      | Where                         | Purpose                                                              |
|-----------------------------------------------------------------------------|-------------------------------|----------------------------------------------------------------------|
| <code>ControlRoom.alarm_thresholds["bp_systolic"] + ["bp_diastolic"]</code> | same                          | New BP threshold keys (adult-norm defaults).                         |
| <code>alarms.silence_alarm(room, alarm_id, duration_s)</code>               | <code>portal/alarms.py</code> | Direct helper for tests + future programmatic callers.               |
| <code>alarms._apply_silenced(room, alarms)</code>                           | same                          | Internal filter that drops cleared alarms + annotates silenced ones. |
| Severity "danger" (rank 4)                                                  | same                          | New tier above critical for dangerous waveforms.                     |

## 5. Limitations

- **No persistence.** `silenced_alarms` lives on the in-memory

ControlRoom — restarts wipe silence state. Acceptable for classroom sessions; if persistence is needed a future M51 could serialize to SQLite.

- **Single silence duration in UI.** Default 120 s; operator can

hit the route directly with `?seconds=N` but the Silence button doesn't expose a duration picker. A future M51 could add a long-press / dropdown for 60s / 120s / 5min.

- **The "danger" WAV is the same as critical.** The 15-asset

library doesn't ship a dedicated danger-priority WAV; we route danger to the HIGH-priority bucket (same as critical). The visual treatment carries the priority difference.

- **\*\*Nurse-station Code Blue confirm dialog is the native browser**

`confirm()`\*\*. A custom modal could match the M39 engage dialog styling, but the native confirm is unmissable and prevents accidental clicks — operator-positive in this context.

- **No audit trail of who silenced.** The `silenced_alarms` map

doesn't record which operator hit Silence. If a class debrief needs that, the route could log an `event_id`. Out of scope.

- **Cleared threshold alarms re-emerge after 24 h.** The cleared

entry's `until` is set to `now + 86400`. If a classroom session runs longer (rare for a teaching scenario), the cleared alarm re-emerges. Acceptable; doc'd.

## 6. Test status

| Test file                                                                                              | Asserts                                                                  | Status | Last run   |
|--------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|--------|------------|
| <code>tests/v7/test_silence_and_code_blue.py::test_silence_a_threshold_alarm_sets_silenced_flag</code> | Silence route works on threshold alarm + sets <code>silenced=True</code> | PASS   | 2026-05-27 |
| <code>...::test_silence_unknown_alarm_id_404</code>                                                    | Unknown alarm id: accept-200 or reject-404                               | PASS   | 2026-05-27 |

| Test file                                                                      | Asserts                                                                                                           | Status | Last run   |
|--------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------|------------|
| ...::test_clear_threshold_alarm_now_works                                      | The bugfix — Clear on threshold returns 200 + alarm gone from feed                                                | PASS   | 2026-05-27 |
| ...::test_clear_device_alarm_still_works                                       | M26 device/scene clear unchanged                                                                                  | PASS   | 2026-05-27 |
| ...::test_bp_systolic_threshold_breach_alarms                                  | sbp=180 with high=130 → threshold alarm                                                                           | PASS   | 2026-05-27 |
| ...::test_bp_diastolic_threshold_breach_alarms                                 | dbp=105 with high=90 → threshold alarm                                                                            | PASS   | 2026-05-27 |
| ...::test_bp_default_thresholds_present                                        | bp_systolic + bp_diastolic in default GET                                                                         | PASS   | 2026-05-27 |
| ...::test_danger_severity_sorts_above_critical                                 | v-fib (danger) appears ABOVE SpO2 (critical) in feed                                                              | PASS   | 2026-05-27 |
| ...::test_danger_severity_maps_to_high_priority_audio                          | severity_to_priority("danger") == "high"; audio_url_for(danger rhythm) → file 04                                  | PASS   | 2026-05-27 |
| ...::test_nurse_code_blue_via_sid_fires_scene                                  | Nurse-station sid (no instructor cookie) fires code.blue at named encounter                                       | PASS   | 2026-05-27 |
| ...::test_nurse_code_blue_rejects_unknown_sid                                  | Bogus sid → 403                                                                                                   | PASS   | 2026-05-27 |
| ...::test_nurse_code_blue_works_with_instructor_cookie                         | Operator cookie path (no nurse_sid) → by="instructor"                                                             | PASS   | 2026-05-27 |
| ...::test_nurse_code_blue_unknown_encounter_404                                | Unknown encounter id → 404                                                                                        | PASS   | 2026-05-27 |
| ...::test_nurse_station_html_carries_bp_threshold_inputs                       | Threshold form has sbp + dbp low/high inputs                                                                      | PASS   | 2026-05-27 |
| ...::test_nurse_station_js_has_silence_handler_and_code_blue_button            | JS source carries data-silence, /silence call, audio.silenced check, ns-code-blue-btn, nurse_code_blue route call | PASS   | 2026-05-27 |
| (regression update)<br>test_threshold_dangerous_rhythm_raises_when_ecg_enabled | Severity assertion critical → danger                                                                              | PASS   | 2026-05-27 |

| Test file     | Asserts                                                 | Status | Last run   |
|---------------|---------------------------------------------------------|--------|------------|
| Full v7 suite | 374 passed, 1 skipped (M20 Playwright skip — unchanged) | PASS   | 2026-05-27 |

## 7. Change list

| Date       | Author      | Change                                                                                                                                                                                                   | Files                                                                                                                                                                                                                                                                         |
|------------|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026-05-27 | claude-code | Initial M50: silence helper + route; clear bugfix for threshold; BP thresholds; danger severity rank 4; nurse-station code-blue route + button; UI updates; 15 new tests + 1 regression assertion update | portal/control_room.py, portal/alarms.py, portal/alarm_sounds.py, portal/server.py, portal/templates/nurse_station.html, portal/static/nurse_station.{js,css}, tests/v7/test_silence_and_code_blue.py (new), tests/v7/test_alarm_thresholds_and_ecg_fix.py (assertion update) |

## 8. Open questions / known issues

- **Persistence.** `silenced_alarms` dict resets on server restart.

If the operator's classroom session restarts mid-flow, silenced alarms wake up. Acceptable for typical teaching sessions; M51 could persist to SQLite.

- **Silence duration UX.** Default 120s; the route accepts any

?seconds=N. A dropdown / long-press for 60s/120s/5min would be nicer.

- **Custom Code Blue confirm modal.** Today uses `window.confirm()`

which is unmissable but doesn't match the M39 dialog styling. Trade-off — defaulted to the safer native confirm.

- **Audit trail.** No record of who silenced/cleared an alarm.

The cohort debrief could surface this if useful. Tracked.

- **No "snooze for 5 min" / smart-snooze.** Standard silence is a

fixed window. Real bedside monitors offer a "Silence Indefinitely" / "Pause Audio" mode for confirmed-critical alarms. Out of scope.

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